

APIs: Groq LLaMA 3.1 + HuggingFace RoBERTa

INN Sight AI

Hotel Review Intelligence Platform

Week 7 — AI API Integration

W7_AIFeatureDemo_TBI-26101076

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GitHub: github.com/mehaksethiii/Stay-INN_sight-AI

1. AI Feature Overview

This submission demonstrates three AI-powered features integrated into INN Sight AI:

Feature 1 — AI Review Analyser (POST /api/ai/analyse)

- User inputs any hotel guest review text
- Backend calls Groq LLaMA 3.1 and HuggingFace RoBERTa in parallel
- Returns: sentiment, confidence, emotion, detected themes, key issues, recommendations, business insight
- Generates a professional AI management response ready to send to the guest
- 3-layer fallback: Groq !' HuggingFace !' Local NLP (never crashes)

Feature 2 — Floating AI Chatbot (POST /api/ai/chat)

- Domain-specific chatbot powered by Groq LLaMA 3.1
- Answers questions about INN Sight AI, dashboard, reviews, AI analysis
- User Input (React Frontend)
- Features: conversation history, markdown rendering, typing indicator
- UI: minimize / maximize / close window controls (macOS-style)
- POST /api/ai/analyse (authenticated, rate-limited)
- Suggested prompts, clear chat, auto-scroll

Feature 3 — AI Sentiment on Review Submit (POST /api/reviews)

- Upgraded existing keyword-matching to real AI classification
- Every new review is classified by Groq LLaMA 3.1 !' stored in MongoDB
- Sentiment (positive/neutral/negative), emotion, themes, in response }

analysis.service.js !' Promise.allSettled([
runGroqAnalysis(), !' Groq LLaMA 3.1
runHFAnalysis() !' HuggingFace RoBERTa
)
})
JSON Response to Frontend !' Displayed in AIAlyser.js

2. Architecture

3. Security

- API keys stored ONLY in backend/.env — never committed to GitHub
- backend/.env listed in .gitignore
- Rate limiting: 20 req/15min (analyse) | 50 req/15min (chat)
- Input sanitization — HTML stripped, prompt injection pattern blocked
- JWT authentication required on all AI endpoints

[Screenshot 2: Loading spinner visible — "Analysing..." button state during API call]

4. Feature Demo Screenshots

[Screenshot 3: Final AI verdict — sentiment banner, themes, recommendations, management response]

The following screenshots demonstrate the AI Review Analyser feature end-to-end on localhost:3000.

Screenshot 1 — User Input Screen

! Shows the AI Analyser input form. User enters guest name and review text, then clicks "Analyse Review".

Screenshot 2 — Loading State

! Shows the loading state with spinner animation while Groq API is processing the review ("H2—5 seconds).

Screenshot 3 — AI Output

! Shows the complete AI output: sentiment label, confidence %, detected emotion, themes, and the AI-generated professional management response with copy button.

5. Network Tab — API Verification

Screenshot 4 — Browser DevTools Network Tab

! Confirms the POST /api/ai/analyse request returns HTTP 200 with AI analysis JSON payload.

6. API Endpoints

POST/
api/ai/
analyse

[Auth]
Dual AI
review
analysis
(Groq + Hu
ggingFace)

POST/
api/ai/
chat

[Auth]
Domain AI
chatbot
powered
by Groq
LLaMA 3.1

POST/
api/
reviews

[Auth]
Submit
review —
AI
sentiment
classificati
on

GET/
api/
reviews

[Open]
Fetch all
reviews
with
sentiment
labels

POST/
api/
auth/
login

[Open]
JWT login

POST/
api/
auth/
google

[Open]
Google

PORT=5000

MONGO_URL=mongodb+srv://<user>:<pass>@cluster.mongodb.net/innsightai

JWT_SECRET=your_jwt_secret_key

FIREBASE_PROJECT_ID=your_firebase_project_id

HUGGINGFACE_API_KEY=your_huggingface_api_key # from huggingface.co/settings/tokens

GROQ_API_KEY=your_groq_api_key # from console.groq.com/keys

The following environment variables are required (set in backend/.env — never committed):

7. Environment Variables

8. Prompt Engineering Summary

Full prompt: "You are a helpful assistant. Please provide a summary of the text below. Summarize: rude."

```
System: You are an expert sentiment analysis model.
```

Prompt: Not a good experience, the rooms were a mess and the staff was incredibly rude.

Output: negative

Better: Understands hotel + vocabulary
Hotel review: the rooms were

```

User: "Hotel Review: 'The rooms were filthy and the staff was incredibly rude.'"
Output: Negative

```

Output: Negative

Prompt variation 3 — Role + Strict Format (Best ')

' Best Prompt: Variation 3 — Used in production (sentiment.service.js)

Why it works best: Role framing activates NLP-specific reasoning. The "ONLY one word" constraint eliminates verbose outputs that break JSON parsing. Domain prefix improves classification of hospitality-specific

phrases. Achieved 95%+ accuracy on test inputs including nuanced cases like "Not terrible for the price" (!

Commit: feat: integrate Grog, LLaMA 3.1 + HuggingFace dual AI analysis, Neutral and "Shockingly good" (1 Positive)

Floating chatbot, AI sentiment on reviews - Week 7 TBI-26101076

Hash: afb555a

Branch: main

9. GitHub Commit

9. GitHub Commit

10. Files Changed (18 files, 1539 insertions)

- backend/routes/ai.js !• AI endpoints (analyse + chat)
- backend/services/groq.service.js !• Groq LLaMA 3.1 caller
- backend/services/huggingface.service.js !• HuggingFace dual model caller
- backend/services/sentiment.service.js !• 3-layer sentiment fallback
- backend/services/analysis.service.js !• Parallel dual AI analysis
- backend/services/chatbot.service.js !• Domain chatbot service
- backend/server.js !• AI sentiment on review submit
- backend/.env.example !• Updated with GROQ_API_KEY
- src/pages/AIAnalyser.js !• Complete AI Analyser UI
- src/components/FloatingChatbot.js !• Chatbot with window controls
- src/App.js !• FloatingChatbot registered globally
- PROMPTS.md !• 3 prompt variations + analysis
- README.md !• Full documentation update